

# CMO.AI Ad & Content Generation Agent

## 1. Introduction

The Ad & Content Generation Agent is a core module within the CMO.AI multi-agent system. It functions as an intelligent content creator that automatically generates high-quality marketing materials for startups and brand-led businesses.

This agent simulates the role of a professional content marketing team by analyzing brand inputs and producing platform-optimized content — personalized through a Retrieval-Augmented Generation (RAG) layer backed by Pinecone vector storage.

## 2. Objective

The primary objective of this agent is to:

- Automatically generate high-quality content for social media posts, email campaigns, and promotional messaging
- Leverage brand-specific knowledge retrieval (RAG) to ensure all content aligns with the brand identity
- Apply platform-specific rules and character limits for Instagram, Twitter, LinkedIn, Facebook, and Email
- Produce multiple content variations to enable A/B testing and audience engagement optimization
- Integrate SEO optimization including keywords, meta descriptions, and suggested titles
- Serve as a modular component within the larger CMO.AI multi-agent architecture

## 3. Input Parameters

The agent accepts the following structured inputs via the ContentRequest schema:

| Parameter    | Type    | Required | Description                                              |
|--------------|---------|----------|----------------------------------------------------------|
| content_type | Literal | Yes      | social_media_post / email_campaign / promotional_message |
| brand_name   | str     | Yes      | Name of the brand or startup                             |

|                 |         |     |                                                   |
|-----------------|---------|-----|---------------------------------------------------|
| industry        | str     | Yes | Industry or business sector                       |
| target_audience | str     | Yes | Description of the intended audience              |
| tone            | Literal | No  | professional / casual / humorous / inspirational  |
| platform        | Literal | No  | instagram / twitter / linkedin / facebook / email |
| topic_or_offer  | str     | Yes | The main topic, product, or offer to promote      |
| cta             | str     | No  | Call to Action text (default: Learn more)         |
| extra_notes     | str     | No  | Any additional brand or content instructions      |

## 4. System Architecture

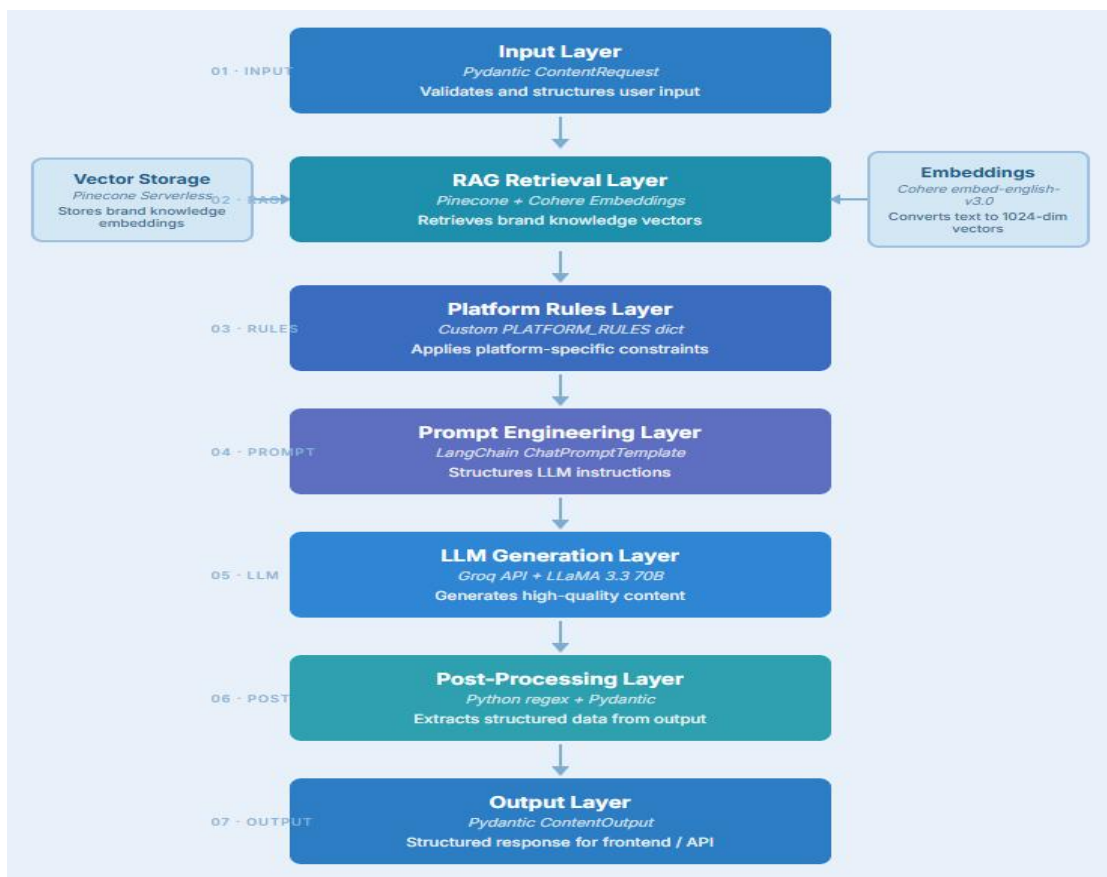
---

The Ad & Content Generation Agent follows a multi-layered architecture that combines Retrieval-Augmented Generation (RAG), platform intelligence, and large language model generation:

### 4.1 Components

- **Input Layer:** Accepts structured ContentRequest objects from the backend API or directly from run\_agent.py
- **RAG Layer (Pinecone + Cohere):** Retrieves brand-specific knowledge including tone guidelines, product info, audience data, and successful post examples
- **Platform Rules Engine:** Applies platform-specific constraints such as character limits, best practices, and optimal posting times
- **Prompt Engineering Layer:** Three specialized ChatPromptTemplates guide the LLM to produce structured, brand-aligned content
- **LLM Layer (Groq + LLaMA 3.3 70B):** Generates the final content using retrieved knowledge and platform rules
- **Post-Processing Layer:** Extracts hashtags, parses content variations, extracts SEO data, and validates character limits
- **Output Layer:** Returns a structured ContentOutput object with all generated content and metadata

## 4.2 Architecture Diagram



## 5. Agent Behavior

The Content Generation Agent demonstrates intelligent, tool-augmented behavior through the following capabilities:

- **Context-aware generation:** Retrieves relevant brand knowledge before generating any content, ensuring all outputs are consistent with the brand identity
- **Platform intelligence:** Automatically selects and applies the correct rules, character limits, and best practices for each social media platform
- **Variation generation:** Produces multiple content variations (A/B testing) for social media posts and promotional messages
- **SEO integration:** Automatically extracts and structures SEO keywords, meta descriptions, and suggested titles from email campaign outputs
- **Validation:** Checks all generated content against platform character limits and reports compliance

## 6. RAG Knowledge Base

The RAG layer is organized around the following knowledge types and retrieval workflow:

### 6.1 Knowledge Base Structure

Each brand stores documents across the following knowledge types:

| Knowledge Type  | Description                      | Example                                                  |
|-----------------|----------------------------------|----------------------------------------------------------|
| tone_guideline  | How the brand communicates       | Warm, friendly, energetic. Speak like a coffee friend.   |
| audience        | Target audience profile          | Young professionals aged 22-35 who value sustainability. |
| product         | Key products or services         | Oat Milk Latte, Cold Brew, Single Origin Pour Over.      |
| brand_values    | Core brand values                | Sustainability, quality sourcing, community, innovation. |
| successful_post | High-performing content examples | Your Monday got an upgrade. Cold Brew hits different.    |

### 6.2 RAG Workflow

- **Step 1 — Embedding:** Brand knowledge documents are embedded using Cohere embed-english-v3.0 (1024 dimensions)
- **Step 2 — Storage:** Embeddings are stored in a Pinecone Serverless index with brand name as metadata filter
- **Step 3 — Retrieval:** At generation time, top-3 most relevant documents are retrieved using cosine similarity search
- **Step 4 — Injection:** Retrieved knowledge is injected into the prompt template under the BRAND KNOWLEDGE BASE section

## 7. Output Structure

The agent returns a structured ContentOutput object containing all generated content and metadata:

| Field             | Type                   | Description                                    |
|-------------------|------------------------|------------------------------------------------|
| content_type      | str                    | Type of content generated                      |
| platform          | str   None             | Target platform                                |
| generated_content | str                    | The full generated content text                |
| variations        | list[ContentVariation] | Parsed A/B variations with IDs                 |
| hashtags          | list[str]              | Extracted hashtags (social media only)         |
| subject_line      | str   None             | Email subject line (email only)                |
| seo               | SEOData   None         | Keywords, meta description, title (email only) |
| platform_rules    | dict   None            | Platform rules that were applied               |
| char_count        | int                    | Character count of generated content           |
| within_limit      | bool                   | Whether content is within platform limit       |

## 8. Technologies Used

| Component       | Technology                 | Version          | Purpose                 |
|-----------------|----------------------------|------------------|-------------------------|
| LLM             | LLaMA 3.3 70B (via Groq)   | groq API         | Content generation      |
| LLM Framework   | LangChain + LangChain-Groq | 0.3.25           | LLM orchestration       |
| Vector Database | Pinecone Serverless        | 6.0.2            | Brand knowledge storage |
| Embeddings      | Cohere embed-english-v3.0  | langchain-cohere | Text vectorization      |
| Data Validation | Pydantic                   | 2.11.4           | Input/output schemas    |
| Environment     | python-dotenv              | 1.1.0            | API key management      |
| Backend Ready   | FastAPI + Uvicorn          | 0.115.12         | API endpoint exposure   |

## 9. Platform-Specific Rules

The agent applies different rules and constraints depending on the target platform:

| Platform  | Char Limit | Best Practices                                                 | Best Post Time          |
|-----------|------------|----------------------------------------------------------------|-------------------------|
| Instagram | 2,200      | Storytelling, emojis, 5-10 hashtags, strong hook in first line | Tue–Fri 9–11AM or 7–9PM |
| Twitter   | 280        | Concise, max 2 hashtags, create curiosity                      | Mon–Thu 8–10AM or 6–9PM |
| LinkedIn  | 3,000      | Hook opener, line breaks, 3-5 hashtags, professional tone      | Tue–Thu 7–9AM or 12–1PM |
| Facebook  | 63,206     | Questions to boost comments, short and engaging                | Wed–Fri 1–4PM           |
| Email     | 5,000      | Strong subject line, single CTA, mobile-friendly               | Tue or Thu 10–11AM      |

---

## 10. Role in CMO.AI System

This agent represents the Content Generation Module within the overall CMO.AI multi-agent architecture. It is designed to be fully modular and integrable with the other agents in the system.

It will be integrated with:

- Brand Coaching Agent — provides brand identity and audience data that seeds the knowledge base
- Marketing Planner and Scheduler Agent — triggers content generation as part of automated campaign schedules
- Image Generation Agent — receives content output to generate matching visual assets
- Video Generation Agent — uses generated copy as scripts for short-form video content
- Performance Analytics Agent — feeds engagement data back to improve future content generation

Together, these agents form a complete end-to-end marketing automation system orchestrated through LangGraph and CrewAI.

---

## 11. Model Selection and Justification

### 11.1 Selected Model

The Ad & Content Generation Agent uses LLaMA 3.3 70B Versatile, accessed through the Groq API via the langchain-groq integration.

## 11.2 Why This Model is Suitable

| Criterion                    | Justification                                                                                            |
|------------------------------|----------------------------------------------------------------------------------------------------------|
| High-Quality Text Generation | Produces professional, brand-aligned marketing content with natural language fluency                     |
| Instruction Following        | Handles complex multi-part prompts including brand knowledge, platform rules, and variation requirements |
| Speed (Groq Infrastructure)  | Groq provides ultra-low latency inference, suitable for real-time user interaction                       |
| Cost-Effective               | Free tier available through Groq API, ideal for academic MVP development                                 |
| No Fine-Tuning Required      | RAG approach provides brand personalization without retraining the model                                 |
| Open Source                  | LLaMA 3.3 is open-source, aligning with the project's academic and research goals                        |

## 11.3 Embedding Model

Cohere embed-english-v3.0 was selected for generating knowledge base embeddings due to its high semantic accuracy, 1024-dimensional output compatible with Pinecone, and free tier availability.

## 12. Testing and Evaluation

The agent was tested across all three content types using the following evaluation criteria:

| Test Case | Content Type        | Platform  | Result                                                          |
|-----------|---------------------|-----------|-----------------------------------------------------------------|
| Test 1    | Social Media Post   | Instagram | Generated 2 variations with 10 hashtags, within 2200 char limit |
| Test 2    | Email Campaign      | Email     | Generated subject line, full body, and SEO section              |
| Test 3    | Promotional Message | General   | Generated 3 variations all under 150 characters                 |

All tests confirmed:

- RAG retrieval successfully injected brand knowledge (audience, products, tone) into generated content
- Platform rules were correctly applied to each content type

- Character count validation worked correctly for all platforms
- SEO data was accurately parsed from email campaign output
- Content variations were correctly identified and structured

## 1- Social Media Post:

|                                                                                       |                                                     |
|---------------------------------------------------------------------------------------|-----------------------------------------------------|
| BRAND NAME                                                                            | INDUSTRY                                            |
| <input type="text" value="FreshBrew"/>                                                | <input type="text" value="Coffee &amp; Beverages"/> |
| TARGET AUDIENCE                                                                       |                                                     |
| <input type="text" value="Young professionals aged 22-35 who love specialty coffee"/> |                                                     |
| TO NE                                                                                 | PLATFORM                                            |
| <input type="text" value="Casual"/>                                                   | <input type="text" value="Instagram"/>              |
| TOPIC / OFFER                                                                         |                                                     |
| <input type="text" value="New oat milk latte launch — 20% off this week"/>            |                                                     |
| CALL TO ACTION                                                                        |                                                     |
| <input type="text" value="Order now via link in bio"/>                                |                                                     |
| <input type="button" value="Generate content ↗"/>                                     |                                                     |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |             |                |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|----------------|
| Social media post                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |             | Instagram      |
| Variation 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Variation 2 |                |
| <p>Hey coffee lovers, we've got a game-changer for you.</p> <p>Introducing our brand new Oat Milk Latte — made with love, rich espresso, and velvety oat milk. To celebrate the launch, we're giving you 20% OFF your first try.</p> <p>Ready to level up your morning routine? Order now via the link in our bio!</p> <p>#FreshBrewVibes #OatMilkLatte #CoffeeLover #NewLaunch #DairyFree #SustainableCoffee #LatteLove #CoffeeObsessed #PlantBased #BrewtifulMornings</p> <p>#FreshBrewVibes #OatMilkLatte #CoffeeLover #NewLaunch #DairyFree #SustainableCoffee #LatteLove #CoffeeObsessed #PlantBased #BrewtifulMornings</p> |             |                |
| CHARACTER COUNT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |             | PLATFORM LIMIT |
| 498 chars                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |             | 2,200 limit    |
| <span>Within limit</span> <span>RAG: brand knowledge applied</span>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |             |                |
| BRAND KNOWLEDGE RETRIEVED FROM PINECONE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |             |                |
| <p><b>audience</b> Young professionals aged 22-35 who value quality, sustainability, and unique coffee experiences.</p> <p><b>product</b> Top products: Oat Milk Latte, Cold Brew, Single Origin Pour Over, and seasonal specials.</p> <p><b>successful_post</b> 'Your Monday just got an upgrade. Cold Brew hits different. Link in bio.' — 3x avg engagement.</p>                                                                                                                                                                                                                                                              |             |                |

## 2- Email Campaign:

|                                                                                       |                                                     |
|---------------------------------------------------------------------------------------|-----------------------------------------------------|
| BRAND NAME                                                                            | INDUSTRY                                            |
| <input type="text" value="FreshBrew"/>                                                | <input type="text" value="Coffee &amp; Beverages"/> |
| TARGET AUDIENCE                                                                       |                                                     |
| <input type="text" value="Young professionals aged 22-35 who love specialty coffee"/> |                                                     |
| TO NE                                                                                 | PLATFORM                                            |
| <input type="text" value="Casual"/>                                                   | <input type="text" value="Email"/>                  |
| TOPIC / OFFER                                                                         |                                                     |
| <input type="text" value="New oat milk latte launch — 20% off this week"/>            |                                                     |
| CALL TO ACTION                                                                        |                                                     |
| <input type="text" value="Order now via link in bio"/>                                |                                                     |
| <input type="button" value="Generate content ↗"/>                                     |                                                     |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Email campaign                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Email |
| <p>Dear Valued Subscriber,</p> <p>We're thrilled to introduce FreshBrew's newest creation: the Oat Milk Latte.</p> <p>Made with our signature single-origin espresso and the creamiest oat milk, it's the perfect blend of bold and smooth — and it's completely plant-based.</p> <p>To celebrate the launch, we're giving you an exclusive 20% off your first order.</p> <p>Here's what makes it special:</p> <ul style="list-style-type: none"><li>— Rich, velvety texture</li><li>— Sustainably sourced oat milk</li><li>— Available hot or iced</li><li>— 100% dairy-free</li></ul> <p>Order now and experience the difference.</p> <p>[ Order Now — 20% Off ]</p> <p>Thank you for being part of the FreshBrew family.</p> <p>Warm regards,<br/>The FreshBrew Team</p> |       |



### 3- Promotional Message:

BRAND NAME

FreshBrew

INDUSTRY

Coffee & Beverages

TARGET AUDIENCE

Young professionals aged 22-35 who love specialty coffee

TONE

Casual

TOPIC / OFFER

New oat milk latte launch — 20% off this week

CALL TO ACTION

Order now via link in bio

Generate content ↗

Promotional message

General

Variation 1

Variation 2

Variation 3

Boost your day! 30% off all FreshBrew for 24hrs only. Shop now!

CHARACTER COUNT

195 chars

PLATFORM LIMIT

2,000 limit

Within limit

RAG: brand knowledge applied

BRAND KNOWLEDGE RETRIEVED FROM PINECONE

audience

Young professionals aged 22-35 who value quality, sustainability, and unique coffee experiences.

product

Top products: Oat Milk Latte, Cold Brew, Single Origin Pour Over, and seasonal specials.

successful\_post

'Your Monday just got an upgrade. Cold Brew hits different. Link in bio.' — 3x avg engagement.

## 13. Conclusion

The Ad & Content Generation Agent successfully demonstrates how generative AI combined with Retrieval-Augmented Generation (RAG), platform intelligence, and structured prompt engineering can produce realistic, high-quality, and brand-consistent marketing content automatically.

The agent is fully proposal-grade, satisfying all requirements outlined in the CMO.AI graduation project proposal including automated content generation, SEO optimization, multi-platform support, content variation testing, and brand knowledge personalization.

It serves as a foundational component in the CMO.AI intelligent marketing automation platform, with all integration points detailed in Section 10.